# Haoyuan Li

https://leehaoyuan.github.io/ | haoyuanl@cs.unc.edu | https://scholar.google.com/citations?hl=en&user=BdfQ44MAAAAJ

## EDUCATION

**University of North Carolina at Chapel Hill, Chapel Hill, US** Sep.2022-May 2027(epected)

***Major: Computer Science, Doctor of Philosophy, Department of Computer Science***

- **Advisor: Snigdha Chaturvedi**

**Harvard University, Cambridge, US** Sep.2020-Jan.2022

***Major: Health Data Science, Master of Science, School of Public Health***

- **Cumulative GPA:** 3.79/4.0

**Renmin University of China, Beijing, CN** Sep.2016–July.2020

***Major: Statistics, Bachelor of Science, School of Statistics***

- **Cumulative GPA:** 3.74/4.0

## INTERNSHIP

**Amazon** May.2025-Aug.2025

- **Position:** Applied Scientist Intern **Mentor:** Zhengyuan Shen

**Microsoft Research Asia** Jan.2021-July.2021

- **Position:** Research Intern **Mentor:** He Di

**Yale University** Jun.2020-Sep.2020

- **Position:** Research Assistant **Advisor:** Dragomir R. Radev

**Stanford University** Jun.2019-Sep.2019

- **Position:** Research Assistant **Advisor:** Ram Rajagopal

## PUBLICATIONS

**Haoyuan Li**, Zhengyuan Shen, Sullam Jeoung, Yueyan Chen, Jiayu Li, Qi Zhu, Shuai Wang, Vassilis N. Ioannidis, Huzefa Rangwala, "BoundRL: Efficient Structured Text Segmentation through Reinforced Boundary Generation", Accepted to ACL 2026 Findings

**Haoyuan Li**, Snigdha Chaturvedi, "Comparative Personalization for Multi-document Summarization", Submitted to COLM 2026

**Haoyuan Li**, Rui Zhang, Snigdha Chaturvedi, 'Improving Fairness of Large Language Models in Multi-document Summarization', ACL 2025

**Haoyuan Li**, Yusen Zhang, Rui Zhang, Snigdha Chaturvedi, 'Coverage-based Fairness in Multi-document Summarization', NAACL 2025 SAC award

**Haoyuan Li,** Snigdha Chaturvedi, 'Rationale-based Opinion Summarization', NAACL 2024

**Haoyuan Li**, Somnath Basu Roy Chowdhury, Snigdha Chaturvedi, 'Aspect-aware Unsupervised Extractive Opinion Summarization', ACL 2023 Findings

Bi, Qiwei, **Haoyuan Li**, and Hanfang Yang. 2021. "Boosting Few-Shot Abstractive Summarization with Auxiliary Tasks.", CIKM 2021

Fabbri, Alexander R., Simeng Han, **Haoyuan Li**, Haoran Li, Marjan Ghazvininejad, Shafiq Joty, Dragomir Radev, and Yashar Mehdad. "Improving Zero and Few-Shot Abstractive Summarization with Intermediate Fine-tuning and Data Augmentation.", NAACL 2021

Wang, Z.*, **Haoyuan Li***, Rajagopal, R., (2019) Urban2Vec: Incorporating Street View Imagery and POIs for Multi-Modal Urban Neighborhood Embedding, AAAI 2020, (*contribute equally)

## REASEARCH & PROJECTS

**Inter-dimension Dependence for Multi-Dimensional Evaluation of Open-Ended Text**

Advisor: Snigdha Chaturvedi, University of North Carolina at Chapel Hill

Summary: We propose CorrGap, a metric that quantifies the extent of interference from unrelated dimensions when using LLM judges for multi-dimensional evaluation on open-ended texts. Using the metric, we show that the interference is pervasive across open-ended text generation tasks and evaluation frameworks. To mitigate the inference, we propose DimCheck, a method that removes unrelated evidence from the Chain-of-Thoughts of LLM judges to make them more focus on target dimensions.

**BoundRL: Efficient Structured Text Segmentation through Reinforced Boundary Generation**

Advisor: Shen Zhengyuan, Amazon

Summary: We propose BoundRL, a novel and efficient approach that jointly performs token-level text segmentation and label prediction for long structured texts. Instead of generating complete contents for each segment, it generates only a sequence of starting tokens, To adapt the model for the output format, BoundRL performs reinforcement learning with verifiable rewards (RLVR) with a reward that jointly optimizes document reconstruction fidelity and semantic alignment. To mitigate entropy collapse, it further constructs intermediate candidates by systematically perturbing a fraction of generated sequences of segments to create stepping stones toward higher-quality solutions.

**Comparative Personalization for Multi-document Summarization**

Advisor: Snigdha Chaturvedi, University of North Carolina at Chapel Hill

Summary: We propose ComPSum, a personalized MDS framework. It first generates a structured analysis of a user by comparing their preferences with other users' preferences. The generated structured analysis is then used to guide the generation of personalized summaries. To evaluate the performance of ComPSum, we propose AuthorMap, a fine-grained reference-free evaluation framework for personalized MDS. It evaluates the personalization of a system based on the authorship attribution

between two personalized summaries generated for different users.

**Improving Fairness of Large Language Models in Multi-document Summarization**

Advisor: Snigdha Chaturvedi, University of North Carolina at Chapel Hill

Summary: To improve the fairness of summaries generated by LLMs, we propose FairPO, a preference tuning method based on DPO. To generate preference pair for tuning, the summaries are generated based on perturbed input document sets. To further improve the corpus-level fairness, we dynamically adjust the weights of each preference pairs.

**Coverage-based Fairness in Multi-Document**

Advisor: Snigdha Chaturvedi, University of North Carolina at Chapel Hill

Summary: To measure the fairness of summaries generated by LLMs, we propose a new summary-level fairness measure, Equal Coverage based on coverage of documents with different social attribute values. Equal Coverage addresses the redundancy issue faced by previous measures like Proportional Representation. We also propose a new corpus-level measure, Coverage Parity, which measures the corpus-level fairness in addition to Equal Coverage.

**Rationale-based Opinion Summarization**

Advisor: Snigdha Chaturvedi, University of North Carolina at Chapel Hill

Summary: To improve the usefulness and interpretability of opinion summaries, we propose a new paradigm, Rationale-based Opinion Summarization, where each represented opinion is paired with a rationale that provides more details. To extract rationales for a representative opinion, we consider four criteria: relatedness, specificity, popularity, and diversity.

**Aspect-aware Unsupervised Extractive Opinion Summarization**

Advisor: Snigdha Chaturvedi, University of North Carolina at Chapel Hill

Summary: To ensure coverage of diverse aspects, we introduce an unsupervised extractive opinion summarization method that automatically identifies the aspects described in the review sentences and then extracts sentences based on their aspects. It identifies the underlying aspects of the review sentences using the roots of noun phrases and adjectives appearing in them. It then extracts review sentences of different aspects to form the summary.

**Boosting few-shot abstractive summarization with auxiliary tasks**

Advisor: Hanfang Yang, Renmin University of China

Summary: To improve the sample efficiency, we develop 3 auxiliary tasks for each training sample in addition to the main task: sentence extraction, object prediction, and triple entailment. We then trained the abstractive summarization model with these tasks in a multi-task learning setting. In this project, I build the BART model for abstractive summarization, use adapters for multi-task learning, run experiments on Xsum, and write the paper.

**Improving Zero and Few-Shot Abstractive Summarization with Intermediate Fine-tuning and Data Augmentation**

Advisor: Dragomir R. Radev, Yale University

Summary: To enable domain adaptation of summarization system with a few labeled samples, we generate pseudo-training pairs from the wiki dataset based on the characteristics of the target dataset such as abstractiveness. We also introduce back-translation and consistency loss to further improve the performance. In the project, I introduce consistency loss between original and augmented documents given pseudo-target summaries and write the paper.

**Urban2Vec: Incorporating Street View Imagery and POIs for Multi-Modal Urban Neighborhood Embedding**

Advisor: Ram Rajagopal, Stanford University

Summary: To facilitate real estate recommendation and demographic surveys, we develop unsupervised methods to generate the embedding of neighborhoods based on the street view image and points of interest. To evaluate the effectiveness of our method, we use the embeddings to predict the demographic features of neighborhoods and run clustering on the embeddings. In the project, I design a way to combine image and text and design methods to evaluate its performance.